In [ ]:

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# a. Import necessary packages
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Input, Dense
from sklearn.model_selection import train_test_split

# b. Load the dataset
data = pd.read_csv('cifar-10.csv')

print(data.head())
print(data.shape)

# Separate labels and image pixels
X = data.drop('label', axis=1).values
y = data['label'].values

# Normalize pixel values
X = X.astype('float32') / 255.0

# Split data into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# c. Define the network architecture
model = Sequential([
    Input(shape=(3072,)),
    Dense(256, activation='relu'),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

# Compile the model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# d. Train the model
history = model.fit(
    X_train, y_train,
    epochs=10,
    batch_size=64,
    validation_split=0.1
)

# e. Evaluate the model
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Loss:", loss)
print("Test Accuracy:", accuracy * 100, "%")

# f. Plot training loss and accuracy
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```
plt.figure(figsize=(12, 5))

# Plot loss
plt.subplot(1, 2, 1)
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Training Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()

# Plot accuracy
plt.subplot(1, 2, 2)
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Training Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.tight_layout()
plt.show()
```

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